

PROJECT SYNOPSIS

1. Title of the Project:

Explainable and Fair AI Decision System

2. Group Members

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3. Introduction

Artificial Intelligence and Machine Learning are widely used for decision-making in Banking, Insurance, Healthcare and Recruitment. However, many ML models do not clearly explain their decisions and may have potential bias. We selected this project to address these issues and improve transparency, fairness and trust in AI-based decisions.

4. Objectives

- To develop an AI-based decision prediction system.
- To explain AI predictions using SHAP and LIME.
- To analyze potential bias and fairness using Fairlearn.
- To provide confidence scores and maintain decision audit records.

5. Problem Statement

Traditional ML models provide predictions but often do not explain why a particular decision was made. Biased data or model behavior may also result in unfair outcomes for different groups. Therefore, a system is needed that provides prediction, explanation, fairness analysis and audit information in one platform.

6. Proposed System / Methodology

The system will accept user/application data, preprocess it using Python, Pandas and NumPy, and generate predictions using Random Forest or XGBoost. SHAP and LIME will explain the predictions, while Fairlearn will analyze fairness. The system will use React.js, FastAPI and PostgreSQL to provide a web-based platform.

System Workflow

User Data → Data Preprocessing → ML Model → Prediction → SHAP/LIME Explanation → Fairness Analysis → Confidence Check → Final Result → Audit Log

Technology Used

Frontend: React.js, HTML, CSS, Tailwind CSS, Axios, Recharts

Backend: Python, FastAPI, REST API

AI/ML: Pandas, NumPy, Scikit-learn, Random Forest/XGBoost, SHAP, LIME, Fairlearn

Database: MySQL

Tools: VS Code, Git, GitHub, Postman

7. Modules / Features

- 1 User/Application Data Management
- 2 Machine Learning Prediction
- 3 SHAP-based Explanation
- 4 LIME-based Local Explanation
- 5 Fairness and Bias Analysis
- 6 Confidence Score
- 7 Human Review Support
- 8 Decision Audit Logging
- 9 AI Decision Dashboard
- 10 Reports and Visualization

8. Expected Outcome

The system will provide AI predictions along with understandable explanations, important feature contributions, confidence scores and fairness analysis. It will also maintain audit records to improve transparency, accountability and trust in AI-based decisions.

9. Conclusion

The Explainable and Fair AI Decision System combines Machine Learning, Explainable AI and Fairness Analysis to make AI decisions more transparent, understandable and responsible. The system can be further extended for applications in Banking, Insurance, Healthcare, Recruitment and other decision-making domains.

10. References

- 1 Scikit-learn Documentation
- 2 SHAP Documentation
- 3 LIME — Local Interpretable Model-Agnostic Explanations
- 4 Fairlearn Documentation
- 5 FastAPI Documentation
- 6 PostgreSQL Documentation